TCCT / TRANSPORT MODE DETECTION

Complete Smartphone Benchmark Results

Nine Sydney V1 XGBoost deployment candidates

Pixel 4a – 29 October 2026

This document exports every numerical aggregate in the benchmark comparison and window-summary tables, with model-size/parity details, complete timing distributions, memory phases, replay measurements, APK accounting, model configuration and separately labelled diagnostic checks. Full-precision aggregates and all per-call/per-snapshot records accompany the document in the data archive.

Primary study: 99 successful fresh-process runs, comprising 45 latency runs, 45 inference-memory runs and nine accelerated replay runs. Latency and memory each have five processes per fitted candidate. The replay study has one process per candidate and is functional validation only.

Across candidates, median native inference ranges from 0.156 to 0.276 ms. Median full Flutter calls range from 0.533 to 0.672 ms. Warmed whole-process PSS ranges from 147.85 to 161.00 MiB.

These figures describe this handset and software configuration. Whole-process memory includes the benchmark harness. There is no independent accuracy score for the full-cohort deployment artifacts.

Units: 1 KiB = 1,024 bytes; 1 MiB = 1,048,576 bytes. Timing tables use microseconds unless explicitly marked otherwise.

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1 Definitions and abbreviations

This reference explains the terms used in the result tables. A statistic must be read together with its units, sample count and table caption: the observations may be inference calls, process medians, memory snapshots or fitted candidates.

1.1 Statistics and table notation

Table 1: Statistical definitions

Term	Definition and interpretation
N / count	Number of observations used for a statistic. Each per-model warm-latency table has N=5,000 calls from five processes; the between-process-median table has N=5. A larger N does not by itself mean more independent models or participants.
Mean	Arithmetic average: sum the observations and divide by N. Unusually slow calls can pull the mean upward.
Median / p50	Middle value of the ordered observations, averaging the two middle values when N is even. Approximately half the observations lie at or below the median.
p95	95th percentile. Approximately 95% of observations are at or below this value and 5% are higher. For latency it describes the slower end of the measured calls. Ties, interpolation and rounding affect the exact proportion.
SD / std	Sample standard deviation: a measure of spread around the mean, in the same units as the observations. This report uses denominator N-1. A larger SD means more dispersion; it is not the standard error or a confidence interval.
N=1, SD=0	For a single observation, sample SD cannot be estimated. The exporter prints zero by convention; this does not demonstrate that the operation has no variability.
Per-model vs window p95	Per-model latency p95 is calculated across 5,000 measured calls. Window-summary p95 is calculated across the three seed candidates' summary values. These percentiles describe different sets of observations.
Confidence interval (CI)	An interval produced by a statistical method to quantify uncertainty in an estimated quantity. A p95 is a distribution percentile, not a 95% confidence interval. This benchmark report does not calculate confidence intervals.
— in a table	No value is available or the metric does not apply. A dash does not represent zero.

Example: a latency p95 of 0.224 ms means approximately 95% of measured calls finished within 0.224 ms. It does not promise that every call finishes within this time. For 5,000 calls, the slowest 5% corresponds to about 250 calls.

The exporter uses linear interpolation at zero-based position 0.95(N-1). Equivalently, order values from x(1) to x(N) using one-based ranks. With h=1+0.95(N-1), j=floor(h), and g=h-j, the percentile is:

$$\text{p95} = (1-g)x_{(j)} + gx_{(\min(j+1,N))}.$$

For N=5,000, h=4,750.05, so the result is 95% of the 4,750th ordered value plus 5% of the 4,751st. For three candidate summaries, h=2.9, so their p95 is 10% of the middle value plus 90% of the largest. Neither calculation is a maximum or a confidence bound.

1.2 Memory measurements

Table 2: Memory definitions

Term	Definition and interpretation
RAM	Random-access memory: the device's working memory. The device RAM figure is distinct from memory attributed to this app.
PSS	Proportional Set Size. Counts memory resident for the process, assigning shared pages in proportion to the number of processes sharing them. It is useful for attributing physical-memory cost without charging every process the full shared page.

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Term	Definition and interpretation
RSS	Resident Set Size. Counts the process's resident memory, including shared pages at their full size. Summing RSS across processes can count the same shared page more than once.
Native PSS	Android's native-heap PSS category, read from Debug.MemoryInfo.nativePss. It is one part of total PSS, not a measurement of every native allocation or all native-library code pages.
Dalvik PSS	Android's managed-runtime heap PSS category, read from Debug.MemoryInfo.dalvikPss. The historical name remains in the API even on Android's ART runtime. It does not represent all Flutter/Dart memory.
Baseline	Snapshot after initializing the global runtime and loading the prepared input/expected fixtures, with no model session loaded. It already includes substantial benchmark-harness memory.
Loaded	Snapshot after model asset reading and inference-session creation, before the first prediction and validation calls.
Warmed	Snapshot after parity validation and, for inference runs, 100 explicit warmup calls. Replay warms the model through validation before playback.
Observed / end	Observed snapshots are taken during inference or replay. End snapshots are taken after that workload; replay end is recorded for each session. These phase labels do not imply continuous memory tracing.
Delta / incremental PSS	A phase snapshot minus the same process's baseline. Deltas are calculated within each process before aggregation. They estimate extra process cost and include runtime/ validation/harness effects, not only model weights.
Observed peak PSS	Largest PSS value among the collected snapshots. A brief higher allocation between snapshots may be missed.
High-water RSS	Largest RSS reached during the process lifetime so far, as reported by the operating system. It includes earlier initialization and validation and is not a per-phase incremental peak.

Illustrative example: if a process has 4 MiB of private resident memory and shares another 10 MiB equally with one other process, its RSS is 14 MiB and its PSS is 4+10/2=9 MiB. Separately, a hypothetical baseline PSS of 120 MiB and warmed PSS of 150 MiB give a within-process delta of 30 MiB. These examples are explanatory, not additional phone measurements.

Memory terminology and category names follow the Android memory inspection documentation and the Debug.MemoryInfo API reference.

1.3 Timing and experimental procedure

Table 3: Timing and benchmark definitions

Term	Definition and interpretation
Inference / prediction	Applying a fitted model to an input feature vector to produce a class label and class probabilities.
Latency	Elapsed time to perform an operation. The native, full Flutter, load and replay timers cover different operations; their captions define the boundary.
Native inference	The timed Java/JNI ONNX Runtime session call using an already constructed input tensor. Tensor creation, output decoding/disposal and Flutter-channel transfer are outside this timer.
Full Flutter call	Elapsed time from requesting inference in Dart until the native result returns, including channel transfer, tensor creation, execution, output decoding and disposal. Feature extraction happens separately.
Load / first inference	Load covers asset reading and session creation with the global runtime already initialized. First inference is the first prediction after loading. Neither guarantees uncached storage reads because prior checksum checks and OS caches can affect them.
Warmup / warm inference	Warmup calls exercise the loaded model before collecting the main timing samples. Warm inference reuses the existing session. This study performs 100 explicit warmups after validation before 1,000 measured calls per inference process.
Batch size 1	One feature vector is classified per inference call. The benchmark does not measure large-batch throughput.

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Term	Definition and interpretation
Process repetition	A separate app-process launch for a candidate. Repeated calls within one process are different from independent process repetitions. The latency and memory studies use five fresh processes per candidate.
Replay	Processing recorded sensor events. Accelerated replay runs without original waiting intervals; real-time replay follows event timestamp spacing. Neither is live sensor acquisition.
Decode / buffer / feature	Decode covers reading/decoding recorded input. Buffer work retains and removes events for overlapping windows. Feature extraction computes the ordered statistical inputs supplied to the model.
Replay processing	Buffer work since the preceding window, quality checks/feature extraction and the full inference call. Decode and waiting are recorded separately or excluded, as specified in the report.
Schedule lag	How late a replay prediction finishes relative to its window's expected completion time. It includes processing and scheduling delays; it is distinct from native inference latency.
Window W / step	W is the duration of sensor history used to compute features. Step is the interval between successive window starts. Here W=100/120/140 seconds and step=50/60/70 seconds.
Observation delay	Time required to collect a full valid sensor window before the first result. It is separate from millisecond-scale processing and may be longer when quality checks reject a window.
Seed / candidate	A seed controls random choices in tuning/training. A candidate is one fitted window/seed model. These candidates have different winning hyperparameters, so seed comparisons do not isolate randomness alone.
Protocol complete	The report's completeness flag: at least five fresh release processes, recorded code versions and no recorded thermal interruptions, plus the required call counts or memory phases. It is not an accuracy guarantee or a claim of population representativeness.

1.4 Units, formats and software abbreviations

Table 4: Units and abbreviations

Term	Definition and interpretation
s / ms / us	Seconds / milliseconds / microseconds. The ASCII suffix us denotes microseconds: 1 second=1,000 ms=1,000,000 us, and 1 ms=1,000 us.
Byte / KiB / MiB	A byte is 8 bits. KiB means kibibyte (1,024 bytes); MiB means mebibyte (1,048,576 bytes). These binary units differ from decimal kB (1,000 bytes) and MB (1,000,000 bytes).
Hz	Hertz, meaning events or cycles per second. In the quality checks it is the sensor sampling rate, not the model prediction rate.
Float32	A 32-bit floating-point numerical representation, used for the model's input features. It does not imply the entire app or all intermediate calculations use 32-bit arithmetic.
CPU / execution provider	Central Processing Unit / the ONNX Runtime backend that executes model operators. This study uses CPUExecutionProvider and does not measure GPU or neural-accelerator performance.
ONNX / ORT	Open Neural Network Exchange / ONNX Runtime. ONNX is the exported model representation; ONNX Runtime is the software that loads and executes it.
APK	Android application package: the .apk file installed on the phone. It includes app/runtime code and bundled models and fixtures, so APK size is different from model graph size.
ABI / ARM64	Application Binary Interface / the 64-bit ARM architecture targeted by this APK. An ABI specifies binary compatibility conventions for the device and native libraries.
SDK / API	Software Development Kit / Application Programming Interface. The Android SDK number recorded here is the platform API level; it is distinct from the device's reported Android release string.
JNI / platform channel	Java Native Interface / Flutter's Dart-to-Android message bridge. JNI connects Java/Kotlin-side runtime calls to native code; the Flutter platform channel adds another layer included in the full-call timer.

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Term	Definition and interpretation
Joblib	Python object-serialization format used for the saved training pipeline. Its byte size includes Python-side pipeline state and is reported separately from the deployed ONNX artifact.
XGBoost	Extreme Gradient Boosting, the boosted-tree classifier family used for all nine candidates.
Macro F1	Unweighted mean of per-class F1 scores. Each F1 combines precision P and recall R as 2PR/(P+R). Scores range from 0 to 1, with larger values better. The saved inner macro F1 values in this document are tuning-selection scores, not independent deployment accuracy.
Parity / rtol / atol	Parity is agreement with the reference implementation. Labels must match exactly; probabilities pass when the absolute difference is no greater than atol + rtol times the absolute reference value. Here atol=1e-6 and rtol=1e-5.
SHA-256 / commit	A content checksum / a recorded Git source revision. Checksums identify the exact input/model bytes; the commit identifies the source version used to build the measured app.

2 Study setup and interpretation

Table 5: Device and fixed execution settings

Item	Recorded value
Device / chipset	Google Pixel 4a / SM7150
Reported Android / SDK	17 / 37
Reported OS fingerprint	google/sunfish/sunfish:13/TQ3A.230805.001.S1/10786265:user/ release-keys
RAM reported by Android (bytes)	5,854,576,640
App / Flutter / ONNX Runtime	1.0.0 / 3.44.0 / 1.24.3
APK source commit	0131eba622252fae8d1f190a5c38150139f0e46a
Build / ABI	Release / ARM64
Provider	CPUExecutionProvider
Batch / intra-op / inter-op threads	1 / 1 / 1
Execution / graph optimization / spinning	Sequential / all / disabled
Explicit inference warmups / measured calls	100 / 1,000 per process
Repetitions / shuffle seed / cooldown	5 / 42 / 2 seconds
Replay sessions / duration	3 sessions, 300 seconds each
Observation windows / steps (seconds)	100, 120, 140 / 50, 60, 70
Input features / classes	11 float32 features / bus=0, car=1, train=2
Recorded thermal interruptions	0

The Android release field reports 17 while the recorded build fingerprint identifies an Android 13 build. Both original values are retained rather than silently resolving the discrepancy.

Table 6: Numerical study inventory

Mode	Processes	Candidates	Warm calls	Replay windows	Memory snapshots
Inference latency	45	9	45,000	0	0
Inference memory	45	9	0	0	630
Accelerated replay	9	9	0	108	0

Table 7: Recorded start/end operating conditions

Field	Observations	Minimum	Maximum
battery level	198	86	87
battery temperature tenths c	198	319	339
screen brightness	198	64	64
thermal status	198	0	0
plugged	198	2	2

Battery temperature is in tenths of a degree Celsius. Plugged=2 means USB power. Screen brightness is the Android setting, not calibrated luminance. Power saving was disabled; the app kept the screen on.

Table 8: Metric boundaries

Item	Recorded value
Native load	Asset read and session creation with the global runtime already initialized.
Native inference	OrtSession.run with a prepared input tensor; excludes tensor creation and output decoding/disposal.
Full Flutter call	Includes platform-channel transfer, tensor construction, native execution, output decoding/disposal and return transfer.
Replay processing	Buffer work since the previous window, feature extraction and full inference call. Decode is recorded separately.

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Item	Recorded value
PSS / RSS	Whole-process proportional/resident memory. Baseline already includes runtime and prepared input/expected fixtures.
Observation delay	Sensor-window duration; distinct from model computation. First prediction requires a full valid window.

Before each run, checksums and Android label/probability parity are verified. The first inference occurs before validation/warmup; measured warm calls follow validation and 100 explicit warmups. Checksum reads can warm filesystem caches: load times are process-start measurements with uncontrolled OS cache, not guaranteed cold-flash reads. Sampled memory peaks can miss brief transients.

Mean and median are ordinary sample summaries. The p95 index is 0.95(N-1), linearly interpolated between adjacent sorted values. SD is the sample standard deviation with denominator N-1; N=1 is reported with SD=0 as a bookkeeping convention, not evidence of zero uncertainty. No confidence intervals are inferred from repeated calls.

3 Compact per-model result summary

Table 9: Core latency, model size and memory results

W s	Seed	ONNX bytes	Native ms	p95 ms	App ms	PSS MiB	Delta MiB
100	42	104,934	0.162	0.224	0.550	155.666	28.911
100	43	186,694	0.207	0.264	0.622	157.185	29.780
100	44	146,737	0.187	0.244	0.589	157.107	29.909
120	42	483,463	0.276	0.348	0.672	161.004	31.216
120	43	180,924	0.202	0.259	0.620	153.504	28.175
120	44	144,588	0.168	0.226	0.551	153.098	26.856
140	42	305,178	0.206	0.269	0.609	151.902	25.460
140	43	129,365	0.167	0.227	0.545	147.854	23.735
140	44	112,978	0.156	0.214	0.533	150.064	26.016

Latency pools 5,000 measured calls per candidate. PSS and delta are medians across five process snapshots. PSS includes Flutter, native runtime, fixtures and validation overhead; delta estimates extra process cost rather than isolated weight storage.

4 Model serialization and numerical parity

Table 10: Artifact sizes, feature-table cases and conversion error

W s	Seed	ONNX bytes	Joblib bytes	Cases	Max abs. error
100	42	104,934	711,619	5,119	$1.19209290 \times 10^{-7}$
100	43	186,694	2,169,332	5,119	$1.19209290 \times 10^{-7}$
100	44	146,737	1,734,904	5,119	$1.19209290 \times 10^{-7}$
120	42	483,463	2,874,958	4,229	$1.19209290 \times 10^{-7}$
120	43	180,924	2,358,712	4,229	$1.19209290 \times 10^{-7}$
120	44	144,588	753,539	4,229	$1.19209290 \times 10^{-7}$
140	42	305,178	1,118,237	3,592	$1.19209290 \times 10^{-7}$
140	43	129,365	932,767	3,592	$1.19209290 \times 10^{-7}$
140	44	112,978	916,069	3,592	$1.19209290 \times 10^{-7}$

Total deployed model bytes: 1,794,861. Total desktop parity cases across nine candidates: 38,820. Exact label agreement is required; probability tolerance is rtol=1e-5 and atol=1e-6. Cases include every saved feature-table row plus one missing-value case per feature and an all-missing case. The same corpus is checked on Android before a successful run is retained.

5 APK size accounting

Table 11: Measured APK ZIP-entry contributions (bytes)

Component	Uncompressed	Stored/compressed
onnx runtime libraries	75,723,476	75,723,476
models	1,794,861	361,910
fixtures	11,929,128	4,511,747
replay	5,856,364	956,009
Entire APK file	—	98,254,773
Other entries and container overhead	—	16,701,631

Native runtime libraries are stored uncompressed. Shared Java code, Flutter libraries, ZIP alignment/signatures and other entries are not attributed entirely to ONNX Runtime. Model graphs are much smaller than the full runtime/package.

6 Run coverage and protocol status

Table 12: All candidate/mode result groups

Window s	Seed	Mode	Runs	ONNX bytes	Complete
100	42	Latency	5	104,934	Yes
100	42	Memory	5	104,934	Yes
100	42	Replay	1	104,934	No
100	43	Latency	5	186,694	Yes
100	43	Memory	5	186,694	Yes
100	43	Replay	1	186,694	No
100	44	Latency	5	146,737	Yes
100	44	Memory	5	146,737	Yes
100	44	Replay	1	146,737	No
120	42	Latency	5	483,463	Yes
120	42	Memory	5	483,463	Yes
120	42	Replay	1	483,463	No
120	43	Latency	5	180,924	Yes
120	43	Memory	5	180,924	Yes
120	43	Replay	1	180,924	No
120	44	Latency	5	144,588	Yes
120	44	Memory	5	144,588	Yes
120	44	Replay	1	144,588	No
140	42	Latency	5	305,178	Yes
140	42	Memory	5	305,178	Yes
140	42	Replay	1	305,178	No
140	43	Latency	5	129,365	Yes
140	43	Memory	5	129,365	Yes
140	43	Replay	1	129,365	No
140	44	Latency	5	112,978	Yes
140	44	Memory	5	112,978	Yes
140	44	Replay	1	112,978	No

Complete is the exporter's protocol-completeness flag. Replay has one process per candidate and is functional validation; latency and inference memory each have five.

Missing candidate IDs: none. Failed/cancelled exports in the successful study report: 0.

7 Inference latency

Table 13: Full Flutter inference call (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5,000	678.060	550.000	1,540.050	526.033
100	43	5,000	764.625	622.000	2,166.400	547.808
100	44	5,000	726.833	589.000	1,709.300	527.701
120	42	5,000	824.628	672.000	1,817.100	508.958
120	43	5,000	763.113	620.000	2,272.000	573.678
120	44	5,000	674.503	551.000	1,508.350	489.587
140	42	5,000	733.131	609.000	1,565.150	499.653
140	43	5,000	676.577	545.000	1,590.350	528.201
140	44	5,000	659.352	533.000	1,951.000	511.776

Table 14: First inference: full Flutter call (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	3,596.800	3,008	5,522.000	1,400.842
100	43	5	3,254.400	3,026	3,966.000	531.712
100	44	5	3,338.600	2,967	4,464.200	828.551
120	42	5	3,312.800	3,115	3,897.000	427.742
120	43	5	3,509.400	2,978	4,730.800	906.962
120	44	5	6,106.600	3,065	13,515.800	5,349.343
140	42	5	3,528	3,882	4,038.000	606.963
140	43	5	4,729.400	5,134	7,081.800	2,013.982
140	44	5	4,554.200	3,772	6,416.200	1,569.320

Table 15: First inference: native execution (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	577.416	575.104	603.208	19.277
100	43	5	722.479	665.833	878.260	115.603
100	44	5	668.729	654.479	730.083	45.456
120	42	5	716.271	736.927	746.271	37.465
120	43	5	639.698	637.604	710.500	55.223
120	44	5	1,151.438	654.636	2,696.928	1,133.716
140	42	5	779.354	715.573	960.334	156.129
140	43	5	612.250	607.292	670.750	46.157
140	44	5	615.167	619.843	671.396	52.351

Table 16: Full Flutter model load (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	16,279.400	15,427	19,785.400	3,044.344
100	43	5	17,686.400	17,650	19,339.200	1,579.125
100	44	5	23,787.400	16,906	39,422.400	11,631.633
120	42	5	35,573.400	30,769	45,047.200	8,316.302
120	43	5	24,545.400	25,447	29,482.200	4,570.567
120	44	5	21,224.800	18,799	32,760.400	8,443.880
140	42	5	31,609.200	32,117	37,271.400	5,144.116
140	43	5	26,163.400	21,279	38,121.400	8,949.551
140	44	5	25,597.800	20,970	37,912.200	9,485.424

Table 17: Native asset read and session creation (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	14,123.053	13,668.959	16,725.783	2,515.532

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Window s	Seed	N	Mean	Median	p95	SD
100	43	5	15,891.314	15,203.335	17,545.586	1,476.086
100	44	5	21,279.419	14,985.522	35,128.701	10,479.548
120	42	5	33,081.170	29,368.076	42,307.900	7,146.542
120	43	5	21,061.815	18,800.992	27,259.034	5,162.773
120	44	5	18,763.283	16,981.356	28,059.242	6,829.948
140	42	5	28,049.169	28,120.993	32,428.534	4,007.222
140	43	5	21,914.325	17,919.221	34,659.660	9,380.262
140	44	5	20,552.190	18,183.440	31,046.889	7,674.486

Table 18: Native inference (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5,000	166.299	162.084	224.067	41.385
100	43	5,000	210.015	207.344	263.859	28.719
100	44	5,000	190.384	187.448	243.649	33.419
120	42	5,000	281.532	275.911	348.437	52.484
120	43	5,000	204.918	201.615	259.480	29.522
120	44	5,000	172.010	167.553	226.469	36.845
140	42	5,000	211.862	206.094	268.854	60.750
140	43	5,000	170.954	166.510	226.927	41.068
140	44	5,000	160.434	156.328	213.541	36.998

Table 19: Between-process native medians (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	160.958	159.323	175.063	11.541
100	43	5	206.312	201.823	219.104	10.255
100	44	5	186.083	185.964	196.995	10.059
120	42	5	275.052	277.265	290.266	16.194
120	43	5	202.120	196.641	217.781	11.716
120	44	5	166.766	166.094	175.568	8.288
140	42	5	206.162	204.870	216.344	9.183
140	43	5	165.661	158.333	182.339	14.098
140	44	5	155.792	152.448	169.578	11.295

N is the number of observations. SD uses the sample standard deviation (N-1); p95 uses linear interpolation. Between-process medians have N=5, rather than N=5,000 inference calls.

8 Inference memory

Table 20: First inference: full Flutter call (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	2,826	2,765	2,972.000	110.236
100	43	5	2,752.600	2,778	2,838.000	92.533
100	44	5	2,997.200	2,940	3,297.200	247.989
120	42	5	3,112.600	2,985	3,614.200	369.063
120	43	5	2,763.800	2,751	2,912.800	140.309
120	44	5	3,371.600	3,216	4,247.800	716.883
140	42	5	2,837.400	2,850	2,931.600	112.855
140	43	5	2,866.800	2,881	3,021.200	140.430
140	44	5	2,811.600	2,751	2,947.600	115.913

Table 21: First inference: native execution (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	628.115	627.968	663.948	30.826
100	43	5	609.177	593.073	654.240	37.141
100	44	5	671.906	655.260	736.865	47.713
120	42	5	711.490	711.667	742.292	28.219
120	43	5	647.854	634.739	716.812	55.300
120	44	5	618.146	622.761	639.843	19.396
140	42	5	660.250	670.417	705.427	43.228
140	43	5	634.073	615.000	696.469	58.256
140	44	5	600.708	609.062	627.865	29.467

Table 22: Full Flutter model load (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	13,002.400	13,123	13,387.400	386.936
100	43	5	15,988	16,085	16,633.400	604.026
100	44	5	14,053.800	14,206	14,358.800	315.554
120	42	5	28,024.400	27,919	28,728.800	522.683
120	43	5	15,792.800	15,539	16,837.800	822.179
120	44	5	14,277.400	14,396	14,862.800	597.025
140	42	5	20,862.800	20,405	22,117.400	1,037.665
140	43	5	13,232.400	13,378	13,501.600	353.846
140	44	5	13,485.600	13,421	14,669.800	1,055.843

Table 23: Native asset read and session creation (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	5	10,787.855	10,783.230	10,987.814	195.443
100	43	5	14,268.314	13,872.553	15,158.366	719.664
100	44	5	12,128.293	12,210.783	12,342.762	211.638
120	42	5	26,054.607	26,083.284	26,617.669	505.924
120	43	5	13,757.908	13,427.970	15,095.096	974.835
120	44	5	12,451.657	12,445.991	13,082.584	484.216
140	42	5	19,026.440	18,556.044	20,271.304	1,006.167
140	43	5	11,541.303	11,620.835	11,701.407	226.843
140	44	5	11,686.699	11,540.314	12,652.949	907.822

Table 24: Published PSS phase medians and within-process changes (KiB)

Window s	Seed	Phase	PSS median	Delta median
100	42	baseline	129,821	0
100	42	loaded	139,037	9,204
100	42	warmed	159,402	29,605
100	42	inference observed	163,295.000	33,342.000
100	42	inference end	164,979	35,158
100	43	baseline	129,887	0
100	43	loaded	139,696	9,796
100	43	warmed	160,957	30,495
100	43	inference observed	163,941.000	34,024.000
100	43	inference end	167,576	36,712
100	44	baseline	129,878	0
100	44	loaded	139,741	9,800
100	44	warmed	160,878	30,627

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Window s	Seed	Phase	PSS median	Delta median
100	44	inference observed	164,359.500	34,385.500
100	44	inference end	166,144	36,277
120	42	baseline	127,965	0
120	42	loaded	142,818	14,648
120	42	warmed	164,868	31,965
120	42	inference observed	165,065.500	35,086.500
120	42	inference end	165,544	37,608
120	43	baseline	128,151	0
120	43	loaded	137,984	9,746
120	43	warmed	157,188	28,851
120	43	inference observed	159,746.000	31,545.500
120	43	inference end	163,629	35,292
120	44	baseline	128,056	0
120	44	loaded	137,286	9,410
120	44	warmed	156,772	27,501
120	44	inference observed	159,618.500	30,671.000
120	44	inference end	163,330	32,676
140	42	baseline	130,942	0
140	42	loaded	139,099	9,604
140	42	warmed	155,548	26,071
140	42	inference observed	158,510.000	28,604.000
140	42	inference end	161,912	32,330
140	43	baseline	127,097	0
140	43	loaded	136,241	9,016
140	43	warmed	151,402	24,305
140	43	inference observed	155,220.000	27,961.000
140	43	inference end	157,491	30,517
140	44	baseline	126,759	0
140	44	loaded	136,144	9,350
140	44	warmed	153,666	26,640
140	44	inference observed	155,248.500	27,505.500
140	44	inference end	157,210	30,077

Each delta is computed against that process's baseline before aggregation. The median delta need not equal the difference of separately pooled medians.

Table 25: Observed maximum whole-process PSS (KiB)

Window s	Seed	Observed peak
100	42	167,984
100	43	169,069
100	44	169,580
120	42	172,322
120	43	164,969
120	44	164,585
140	42	165,068
140	43	158,485
140	44	160,818

9 Accelerated replay

Table 26: Full Flutter model load (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	1	13,617	13,617	13,617.000	0
100	43	1	18,075	18,075	18,075.000	0
100	44	1	33,497	33,497	33,497.000	0
120	42	1	29,348	29,348	29,348.000	0
120	43	1	15,841	15,841	15,841.000	0
120	44	1	40,989	40,989	40,989.000	0
140	42	1	22,081	22,081	22,081.000	0
140	43	1	13,523	13,523	13,523.000	0
140	44	1	16,759	16,759	16,759.000	0

Table 27: Native asset read and session creation (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	1	11,855.522	11,855.522	11,855.522	0
100	43	1	16,198.648	16,198.648	16,198.648	0
100	44	1	31,844.951	31,844.951	31,844.951	0
120	42	1	26,466.304	26,466.304	26,466.304	0
120	43	1	13,333.908	13,333.908	13,333.908	0
120	44	1	39,441.774	39,441.774	39,441.774	0
140	42	1	20,488.336	20,488.336	20,488.336	0
140	43	1	12,065.886	12,065.886	12,065.886	0
140	44	1	14,720.366	14,720.366	14,720.366	0

Table 28: Accelerated replay processing (microseconds)

Window s	Seed	N	Mean	Median	p95	SD
100	42	15	19,993.800	19,126	28,755.600	5,112.657
100	43	15	20,480.733	18,837	30,894.200	5,286.460
100	44	15	20,543.133	18,676	29,264.600	5,211.746
120	42	12	23,846.500	23,067.500	33,120.400	5,655.736
120	43	12	23,671.583	21,558.500	37,275.150	7,062.194
120	44	12	23,404	21,029.500	34,948.400	6,281.952
140	42	9	27,905.778	27,335	34,318.000	4,834.753
140	43	9	27,727.222	27,409	35,679.600	5,611.867
140	44	9	29,349	30,548	39,572.600	6,920.395

10 Window summaries across three fitted candidates

These summaries weight each candidate equally. The three candidates at a window have seeds 42, 43 and 44 and different winning hyperparameters. A p95 in this section is across three candidate summaries, not a tail inference latency or confidence bound.

Table 29: Window-group membership

Mode	Window s	Candidates	Seeds
Latency	100	3	42, 43, 44
Latency	120	3	42, 43, 44
Latency	140	3	42, 43, 44
Memory	100	3	42, 43, 44
Memory	120	3	42, 43, 44
Memory	140	3	42, 43, 44
Replay	100	3	42, 43, 44

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Mode	Window s	Candidates	Seeds
Replay	120	3	42, 43, 44
Replay	140	3	42, 43, 44

Table 30: Candidate-level distribution: app us median

Mode	Window s	N	Mean	Median	p95	SD
Latency	100	3	587.000	589.000	618.700	36.042
Latency	120	3	614.333	620.000	666.800	60.699
Latency	140	3	562.333	545.000	602.600	40.857

Table 31: Candidate-level distribution: model bytes

Mode	Window s	N	Mean	Median	p95	SD
Latency	100	3	146,121.667	146,737	182,698.300	40,883.473
Latency	120	3	269,658.333	180,924	453,209.100	186,049.463
Latency	140	3	182,507	129,365	287,596.700	106,551.697
Memory	100	3	146,121.667	146,737	182,698.300	40,883.473
Memory	120	3	269,658.333	180,924	453,209.100	186,049.463
Memory	140	3	182,507	129,365	287,596.700	106,551.697
Replay	100	3	146,121.667	146,737	182,698.300	40,883.473
Replay	120	3	269,658.333	180,924	453,209.100	186,049.463
Replay	140	3	182,507	129,365	287,596.700	106,551.697

Table 32: Candidate-level distribution: native us median

Mode	Window s	N	Mean	Median	p95	SD
Latency	100	3	185.625	187.448	205.354	22.685
Latency	120	3	215.026	201.615	268.481	55.410
Latency	140	3	176.311	166.510	202.136	26.291

Table 33: Candidate-level distribution: observed peak pss kib

Mode	Window s	N	Mean	Median	p95	SD
Memory	100	3	168,877.667	169,069	169,528.900	815.022
Memory	120	3	167,292	164,969	171,586.700	4,360.337
Memory	140	3	161,457	160,818	164,643.000	3,337.696

Table 34: Candidate-level distribution: replay processing us median

Mode	Window s	N	Mean	Median	p95	SD
Replay	100	3	18,879.667	18,837	19,097.100	228.014
Replay	120	3	21,885.167	21,558.500	22,916.600	1,057.542
Replay	140	3	28,430.667	27,409	30,234.100	1,834.038

11 Additional snapshot distributions

11.1 Inference-memory study

Table 35: Total PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	5	129,848.200	129,821	130,085.000	184.385
100	42	inference end	5	165,336.600	164,979	166,532.400	1,084.030
100	42	observed	50	163,250.280	163,295.000	167,227.800	2,411.709
100	42	loaded	5	139,079.800	139,037	139,268.000	146.057
100	42	warmed	5	160,002.600	159,402	162,907.400	2,129.009
100	43	baseline	5	130,135	129,887	131,287.800	845.240
100	43	inference end	5	167,082.600	167,576	168,270.000	1,397.846
100	43	observed	50	164,115.760	163,941.000	167,774.850	2,265.076
100	43	loaded	5	139,623.600	139,696	139,809.600	237.536
100	43	warmed	5	160,520.800	160,957	161,315.800	1,038.267
100	44	baseline	5	129,861.600	129,878	130,194.400	328.314
100	44	inference end	5	166,252.600	166,144	168,117.400	1,618.183
100	44	observed	50	164,234.440	164,359.500	168,183.950	2,466.138
100	44	loaded	5	139,449.400	139,741	139,802.400	618.965
100	44	warmed	5	161,294	160,878	163,970.400	2,044.042
120	42	baseline	5	130,144.200	127,965	135,181.400	3,733.821
120	42	inference end	5	167,894.800	165,544	172,059.000	3,475.860
120	42	observed	50	165,274.140	165,065.500	169,888.000	3,149.060
120	42	loaded	5	143,962.600	142,818	146,372.600	2,122.098
120	42	warmed	5	162,823.800	164,868	165,809.600	3,375.560
120	43	baseline	5	128,147.200	128,151	128,338.600	193.309
120	43	inference end	5	162,919.800	163,629	164,804.200	1,881.399
120	43	observed	50	159,637.240	159,746.000	163,642.950	2,439.611
120	43	loaded	5	137,958.200	137,984	138,084.400	169.315
120	43	warmed	5	156,289.600	157,188	157,870.400	1,684.981
120	44	baseline	5	128,898.800	128,056	131,622.800	2,054.066
120	44	inference end	5	162,094.400	163,330	164,136.000	2,325.786
120	44	observed	50	159,486.400	159,618.500	163,175.500	2,480.866
120	44	loaded	5	137,622.600	137,286	139,324.200	1,279.658
120	44	warmed	5	156,520.600	156,772	158,091.000	1,809.183
140	42	baseline	5	130,005	130,942	132,388.600	2,307.688
140	42	inference end	5	161,911.600	161,912	164,466.200	1,965.329
140	42	observed	50	158,613.680	158,510.000	162,285.400	2,175.763
140	42	loaded	5	139,732.400	139,099	141,756.600	1,479.734
140	42	warmed	5	155,998.200	155,548	158,342.600	1,940.834
140	43	baseline	5	127,256.400	127,097	128,009.800	556.214
140	43	inference end	5	157,539.400	157,491	158,001.400	394.021
140	43	observed	50	154,998.920	155,220.000	157,933.400	1,887.014
140	43	loaded	5	136,242	136,241	136,473.800	210.426
140	43	warmed	5	152,180	151,402	153,960.600	1,540.811
140	44	baseline	5	127,834.200	126,759	130,241.600	1,850.067
140	44	inference end	5	157,267.400	157,210	157,920.800	495.309
140	44	observed	50	155,184.200	155,248.500	158,160.200	2,000.481
140	44	loaded	5	136,440.600	136,144	137,861.400	1,047.815
140	44	warmed	5	153,242.400	153,666	153,931.400	1,048.779

Table 36: Resident set size by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	5	195,733.600	195,712	195,907.200	142.761
100	42	inference end	5	233,837.600	233,640	235,052.800	1,188.783
100	42	observed	50	231,527.280	231,318.000	235,225.400	2,452.851
100	42	loaded	5	204,969.600	204,888	205,202.400	211.596
100	42	warmed	5	228,238.400	227,728	230,964.000	2,014.439
100	43	baseline	5	195,266.400	195,340	195,528.000	276.443

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Window s	Seed	Phase	N	Mean	Median	p95	SD
100	43	inference end	5	235,356	236,092	236,441.600	1,358.311
100	43	observed	50	232,153.360	232,076.000	236,147.000	2,434.399
100	43	loaded	5	205,273.600	205,368	205,562.400	317.157
100	43	warmed	5	228,492.800	228,484	229,455.200	961.592
100	44	baseline	5	195,403.200	195,624	196,093.600	628.544
100	44	inference end	5	234,744	234,904	236,975.200	1,996.535
100	44	observed	50	232,496.080	232,290.000	236,316.800	2,551.010
100	44	loaded	5	205,155.200	205,288	205,694.400	541.061
100	44	warmed	5	229,387.200	229,320	231,524.800	1,829.630
120	42	baseline	5	196,992	194,328	202,116.800	4,169.351
120	42	inference end	5	238,160.800	234,328	245,024.000	5,551.244
120	42	observed	50	235,299.680	234,898.000	242,227.400	4,513.787
120	42	loaded	5	211,490.400	209,140	216,151.200	3,898.186
120	42	warmed	5	232,773.600	233,352	238,292.800	4,990.459
120	43	baseline	5	193,990.400	194,092	194,516.000	571.584
120	43	inference end	5	231,311.200	231,532	233,108.000	1,607.964
120	43	observed	50	227,873.200	227,968.000	232,164.000	2,619.493
120	43	loaded	5	203,755.200	204,012	204,178.400	632.692
120	43	warmed	5	224,365.600	224,340	226,014.400	1,482.103
120	44	baseline	5	193,734.400	193,564	194,676.000	712.413
120	44	inference end	5	230,542.400	231,752	232,619.200	2,222.486
120	44	observed	50	227,796.640	227,866.000	231,597.200	2,555.680
120	44	loaded	5	203,325.600	203,192	204,574.400	955.337
120	44	warmed	5	224,676.800	224,964	226,171.200	1,656.559
140	42	baseline	5	194,178.400	192,872	198,244.000	3,016.144
140	42	inference end	5	231,368.800	230,660	235,877.600	3,330.220
140	42	observed	50	227,882.240	227,220.000	234,363.400	3,256.135
140	42	loaded	5	206,502.400	205,348	210,488.800	2,945.601
140	42	warmed	5	225,172	223,868	229,566.400	3,256.757
140	43	baseline	5	192,682.400	192,708	192,980.800	258.404
140	43	inference end	5	226,241.600	226,128	226,770.400	403.838
140	43	observed	50	223,428.880	223,514.000	226,363.000	1,910.133
140	43	loaded	5	202,380.800	202,384	202,572.800	169.856
140	43	warmed	5	220,626.400	219,744	222,409.600	1,596.019
140	44	baseline	5	192,667.200	191,992	194,338.400	1,208.806
140	44	inference end	5	225,523.200	225,436	226,100.000	542.206
140	44	observed	50	223,254.960	223,262.000	226,537.200	2,139.354
140	44	loaded	5	202,024.800	201,516	203,483.200	1,059.673
140	44	warmed	5	221,140.800	221,276	222,312.000	1,190.225

Table 37: Native PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	5	23,146	23,178	23,232.400	88.679
100	42	inference end	5	26,310	26,294	26,379.600	58.447
100	42	observed	50	26,303.680	26,300.000	26,430.000	67.356
100	42	loaded	5	26,052.400	26,030	26,128.400	59.840
100	42	warmed	5	26,298.800	26,290	26,387.600	73.561
100	43	baseline	5	23,322.800	23,406	23,459.600	155.959
100	43	inference end	5	26,856.400	26,874	26,896.400	38.843
100	43	observed	50	26,815.600	26,820.000	26,896.200	53.250
100	43	loaded	5	26,691.600	26,766	26,832.400	162.742
100	43	warmed	5	26,838.800	26,834	26,903.600	57.786
100	44	baseline	5	23,224.400	23,206	23,290.800	52.581
100	44	inference end	5	26,731.600	26,846	26,874.800	259.594
100	44	observed	50	26,689.520	26,798.000	26,862.800	241.706

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Window s	Seed	Phase	N	Mean	Median	p95	SD
100	44	loaded	5	26,503.600	26,626	26,680.400	307.839
100	44	warmed	5	26,717.200	26,834	26,866.800	261.479
120	42	baseline	5	23,101.200	23,030	23,426.800	247.425
120	42	inference end	5	30,294	30,310	30,464.400	153.597
120	42	observed	50	30,267.200	30,262.000	30,496.600	142.023
120	42	loaded	5	30,407.600	30,342	30,571.600	120.983
120	42	warmed	5	30,361.200	30,314	30,556.400	148.200
120	43	baseline	5	22,878.800	22,878	22,947.600	65.385
120	43	inference end	5	26,433.200	26,446	26,469.200	39.233
120	43	observed	50	26,395.840	26,414.000	26,458.400	47.920
120	43	loaded	5	26,177.200	26,198	26,210.800	38.095
120	43	warmed	5	26,392.400	26,410	26,453.200	56.043
120	44	baseline	5	22,907.600	22,950	23,038.000	121.016
120	44	inference end	5	26,148.400	25,890	26,592.400	382.995
120	44	observed	50	26,132.320	25,878.000	26,590.000	348.559
120	44	loaded	5	25,946	25,726	26,363.600	365.108
120	44	warmed	5	26,130	25,862	26,572.400	401.995
140	42	baseline	5	22,886.800	22,802	23,182.800	217.208
140	42	inference end	5	28,642	28,646	28,727.600	90.907
140	42	observed	50	28,622.480	28,634.000	28,714.000	76.480
140	42	loaded	5	28,515.600	28,526	28,613.200	97.020
140	42	warmed	5	28,598.800	28,606	28,689.200	98.241
140	43	baseline	5	22,774	22,754	22,903.600	111.104
140	43	inference end	5	26,194	26,214	26,252.400	53.516
140	43	observed	50	26,152.880	26,154.000	26,234.000	53.128
140	43	loaded	5	26,078.800	26,094	26,114.800	38.616
140	43	warmed	5	26,149.200	26,162	26,181.200	30.252
140	44	baseline	5	22,769.200	22,838	22,880.400	134.206
140	44	inference end	5	25,935.600	25,926	25,977.200	40.457
140	44	observed	50	25,895.440	25,894.000	25,960.200	46.541
140	44	loaded	5	25,742	25,746	25,813.200	55.929
140	44	warmed	5	25,907.600	25,930	25,943.600	40.457

Table 38: Dalvik PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	5	1,223.200	1,224	1,228.000	5.215
100	42	inference end	5	24,577.600	24,580	24,580.000	3.578
100	42	observed	50	22,036.800	22,048.000	24,120.000	1,352.018
100	42	loaded	5	1,473.600	1,476	1,476.000	3.578
100	42	warmed	5	19,888.800	19,888	19,892.000	3.347
100	43	baseline	5	1,220	1,220	1,224.000	4.000
100	43	inference end	5	24,735.200	24,736	24,743.200	7.155
100	43	observed	50	22,195.680	22,210.000	24,276.400	1,351.887
100	43	loaded	5	1,631.200	1,632	1,636.000	5.215
100	43	warmed	5	20,047.200	20,048	20,055.200	7.155
100	44	baseline	5	1,219.200	1,220	1,223.200	3.347
100	44	inference end	5	24,654.400	24,656	24,656.000	2.191
100	44	observed	50	22,115.120	22,128.000	24,198.200	1,351.946
100	44	loaded	5	1,552	1,552	1,552.000	0.000
100	44	warmed	5	19,966.400	19,968	19,968.000	2.191
120	42	baseline	5	1,223.200	1,224	1,227.200	3.347
120	42	inference end	5	22,213.600	22,212	22,216.000	2.191
120	42	observed	50	19,674.320	19,688.000	21,756.000	1,353.095
120	42	loaded	5	2,216.800	2,216	2,219.200	1.789
120	42	warmed	5	17,522.400	17,520	17,527.200	3.578

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Window s	Seed	Phase	N	Mean	Median	p95	SD
120	43	baseline	5	1,220.800	1,220	1,223.200	1.789
120	43	inference end	5	21,620	21,620	21,620.000	0.000
120	43	observed	50	19,079.760	19,094.000	21,162.200	1,352.924
120	43	loaded	5	1,624	1,624	1,624.000	0.000
120	43	warmed	5	16,928	16,928	16,928.000	0.000
120	44	baseline	5	1,219.200	1,220	1,223.200	3.347
120	44	inference end	5	21,547.200	21,548	21,551.200	3.347
120	44	observed	50	19,007.680	19,022.000	21,088.000	1,352.884
120	44	loaded	5	1,551.200	1,552	1,555.200	3.347
120	44	warmed	5	16,856	16,856	16,859.200	2.828
140	42	baseline	5	1,220.800	1,220	1,227.200	5.933
140	42	inference end	5	19,636	19,636	19,639.200	2.828
140	42	observed	50	17,094.560	17,110.000	19,180.000	1,352.638
140	42	loaded	5	1,863.200	1,864	1,867.200	4.382
140	42	warmed	5	14,940	14,940	14,943.200	2.828
140	43	baseline	5	1,220	1,220	1,224.000	4.000
140	43	inference end	5	19,294.400	19,292	19,305.600	8.295
140	43	observed	50	16,753.520	16,770.000	18,834.200	1,352.612
140	43	loaded	5	1,519.200	1,520	1,523.200	3.347
140	43	warmed	5	14,600	14,596	14,610.400	8.000
140	44	baseline	5	1,219.200	1,220	1,223.200	3.347
140	44	inference end	5	19,259.200	19,260	19,263.200	3.347
140	44	observed	50	16,717.680	16,732.000	18,802.200	1,352.621
140	44	loaded	5	1,488	1,488	1,491.200	2.828
140	44	warmed	5	14,564	14,564	14,567.200	2.828

Resident-set high-water values are lifetime process maxima and include earlier validation. They are not instantaneous incremental model peaks.

Table 39: Lifetime high-water RSS extrema observed (KiB)

Window s	Seed	Snapshots	Minimum	Maximum
100	42	70	197,768	237,744
100	43	70	197,092	237,864
100	44	70	195,816	237,896
120	42	70	196,964	245,620
120	43	70	196,048	233,764
120	44	70	194,856	233,872
140	42	70	196,100	237,144
140	43	70	196,156	227,344
140	44	70	194,728	229,324

12 Accelerated replay component timings

Table 40: Replay window coverage and largest retained row count

W s	Seed	Windows	Accepted	Rejected	Max rows
100	42	15	15	0	6,696
100	43	15	15	0	6,696
100	44	15	15	0	6,696
120	42	12	12	0	7,719
120	43	12	12	0	7,719
120	44	12	12	0	7,719
140	42	9	9	0	8,742

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W s	Seed	Windows	Accepted	Rejected	Max rows
140	43	9	9	0	8,742
140	44	9	9	0	8,742

All replay sessions are streamed with bounded time buffers. Their selected prefixes are 300 seconds long; prediction steps are half the window duration. Quality requires accelerometer/gyroscope/magnetometer rates of at least 30/4/2 Hz and gaps no larger than 500 ms, including window boundaries. Quality counts duplicate rows, but statistics use the first finite vector at each timestamp. Population SD and original m/s-squared, rad/s and microtesla units are preserved.

Table 41: Replay component: decode us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	20,972.800	15,523	37,262.100	9,338.630
100	43	15	22,933.933	18,409	40,405.900	10,201.231
100	44	15	21,019.600	16,359	38,622.700	9,460.743
120	42	12	28,947.083	26,875.000	44,195.700	10,847.405
120	43	12	28,962.833	27,134.500	45,700.400	10,747.095
120	44	12	29,140.500	24,471.500	49,763.150	13,290.697
140	42	9	34,785.778	34,644	53,905.400	13,328.968
140	43	9	33,045	35,848	51,492.800	13,579.893
140	44	9	33,419.778	23,640	57,888.400	15,872.357

Table 42: Replay component: buffer us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	449.867	404	604.600	93.783
100	43	15	486.133	440	741.100	132.833
100	44	15	440.467	423	554.800	74.717
120	42	12	541	533.500	670.450	89.678
120	43	12	500.667	481.000	618.600	68.387
120	44	12	538.750	558.500	615.150	74.108
140	42	9	726	588	1,231.400	288.509
140	43	9	661	557	1,029.000	221.615
140	44	9	665	607	1,005.200	211.089

Table 43: Replay component: feature us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	15,048	13,932	18,864.900	2,607.257
100	43	15	15,209.867	15,342	18,640.000	2,355.675
100	44	15	14,932.400	14,130	20,116.000	2,598.380
120	42	12	17,662.750	16,532.500	24,136.100	3,234.061
120	43	12	17,454.167	16,730.500	22,044.300	2,691.271
120	44	12	17,479.583	16,667.000	22,662.350	2,971.220
140	42	9	23,103.778	22,153	27,929.800	3,808.245
140	43	9	22,690.889	22,007	27,654.800	4,082.983
140	44	9	22,019.556	21,030	27,790.200	4,070.198

Table 44: Replay component: processing us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	19,993.800	19,126	28,755.600	5,112.657
100	43	15	20,480.733	18,837	30,894.200	5,286.460
100	44	15	20,543.133	18,676	29,264.600	5,211.746

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W s	Seed	N	Mean	Median	p95	SD
120	42	12	23,846.500	23,067.500	33,120.400	5,655.736
120	43	12	23,671.583	21,558.500	37,275.150	7,062.194
120	44	12	23,404	21,029.500	34,948.400	6,281.952
140	42	9	27,905.778	27,335	34,318.000	4,834.753
140	43	9	27,727.222	27,409	35,679.600	5,611.867
140	44	9	29,349	30,548	39,572.600	6,920.395

Table 45: Replay component: prediction.native us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	1,575.920	1,899.167	2,776.510	918.205
100	43	15	2,592.955	1,998.698	7,159.375	3,792.199
100	44	15	3,612.080	1,900.209	9,435.308	4,177.485
120	42	12	3,434.002	1,968.255	11,170.123	4,083.885
120	43	12	4,337.053	1,956.719	18,639.023	7,021.152
120	44	12	1,879.358	1,883.776	3,613.683	1,221.802
140	42	9	1,583.148	1,908.698	2,103.615	749.902
140	43	9	1,648.814	1,905.104	2,515.156	854.353
140	44	9	3,093.386	1,863.542	11,003.262	5,203.740

Table 46: Replay component: prediction.app us (microseconds)

W s	Seed	N	Mean	Median	p95	SD
100	42	15	4,495.933	3,128	11,910.400	4,479.425
100	43	15	4,784.733	3,158	14,012.400	4,806.463
100	44	15	5,170.267	3,143	10,970.900	4,286.111
120	42	12	5,642.750	3,202.500	14,719.450	4,923.310
120	43	12	5,716.750	3,181.000	20,860.450	7,289.010
120	44	12	5,385.667	3,160.000	16,599.750	5,452.801
140	42	9	4,076	3,175	9,263.000	3,406.263
140	43	9	4,375.333	3,186	11,096.800	4,454.732
140	44	9	6,664.444	3,230	16,781.600	5,728.154

Replay processing in the measured APK used per-event integer-microsecond rounding for buffering; it can undercount sub-microsecond event work. The current source accumulates high-resolution elapsed ticks. The earlier replay metadata also stated inference-only defaults of 100 warmups/1,000 calls; those counters do not describe replay. Classification/feature parity was unaffected. Later diagnostics of the corrected APK are separate below.

13 Separately recorded diagnostic results

13.1 Initial corrected-release latency smoke check

Diagnostic results below are kept separate from the 99-run study. They do not satisfy a five-process benchmark protocol.

Model: sydney-v1-w100-s42; mode: latency; status: complete.

Source APK commit: 0131eba622252fae8d1f190a5c38150139f0e46a. Result time (UTC): 2026-10-29T06:25:37.171461Z.

Table 47: Diagnostic timing distributions (microseconds)

Operation	Metric	N	Mean	Median	p95	SD
first	app	1	4,276	4,276	4,276.000	0
first	native	1	635.625	635.625	635.625	0

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Operation	Metric	N	Mean	Median	p95	SD
load	app	1	15,580	15,580	15,580.000	0
load	native	1	14,151.616	14,151.616	14,151.616	0
warm	app	1,000	658.903	526.000	1,208.350	545.984
warm	native	1,000	177.253	175.521	228.763	28.815

13.2 Full single-model replay memory check

Diagnostic results below are kept separate from the 99-run study. They do not satisfy a five-process benchmark protocol.

Model: sydney-v1-w100-s42; mode: replay_fast_memory; status: complete.

Source APK commit: 0131eba622252fae8d1f190a5c38150139f0e46a. Result time (UTC): 2026-10-29T06:35:42.688459Z.

Table 48: Diagnostic timing distributions (microseconds)

Operation	Metric	N	Mean	Median	p95	SD
load	app	1	12,146	12,146	12,146.000	0
load	native	1	10,724.272	10,724.272	10,724.272	0
replay	buffer	15	470.467	416	727.600	169.746
replay	decode	15	23,179.200	20,358	40,766.800	10,661.544
replay	feature	15	15,423.333	14,579	20,592.400	2,733.849
replay	processing	15	20,612	19,236	32,867.700	6,293.215
replay prediction	app	15	4,718.200	2,995	15,731.100	5,214.755
replay prediction	native	15	2,522.091	1,952.552	7,161.521	4,549.628

13.3 Diagnostic process memory

Table 49: Total PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	128,401	128,401	128,401.000	0
100	42	loaded	1	137,701	137,701	137,701.000	0
100	42	replay end	3	178,864.333	182,654	198,922.400	23,986.091
100	42	replay observed	3	177,236.667	183,530	186,575.600	13,934.113
100	42	warmed	1	160,318	160,318	160,318.000	0

Table 50: Resident set size by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	193,984	193,984	193,984.000	0
100	42	loaded	1	203,440	203,440	203,440.000	0
100	42	replay end	3	241,445.333	250,732	253,745.200	19,058.093
100	42	replay observed	3	245,157.333	251,512	254,402.800	13,881.494
100	42	warmed	1	228,380	228,380	228,380.000	0

Table 51: Native PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	23,354	23,354	23,354.000	0
100	42	loaded	1	26,214	26,214	26,214.000	0
100	42	replay end	3	27,124.333	27,170	27,172.700	81.709
100	42	replay observed	3	26,823.333	27,058	27,090.400	438.002

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Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	warmed	1	26,294	26,294	26,294.000	0

Table 52: Dalvik PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	1,220	1,220	1,220.000	0
100	42	loaded	1	1,472	1,472	1,472.000	0
100	42	replay end	3	26,233.333	30,588	39,973.200	17,374.232
100	42	replay observed	3	30,697.333	30,940	40,321.600	10,790.047
100	42	warmed	1	19,428	19,428	19,428.000	0

Resident-set high-water values are lifetime process maxima and include earlier validation. They are not instantaneous incremental model peaks.

Table 53: Lifetime high-water RSS extrema observed (KiB)

Window s	Seed	Snapshots	Minimum	Maximum
100	42	9	197,508	274,056

13.4 Real-time pacing and deliberate cancellation

Diagnostic results below are kept separate from the 99-run study. They do not satisfy a five-process benchmark protocol.

Model: sydney-v1-w100-s42; mode: replay_realtime; status: cancelled.

Source APK commit: 0131eba622252fae8d1f190a5c38150139f0e46a. Result time (UTC): 2026-10-29T06:38:31.686864Z.

Table 54: Diagnostic timing distributions (microseconds)

Operation	Metric	N	Mean	Median	p95	SD
load	app	1	25,756	25,756	25,756.000	0
load	native	1	17,163.283	17,163.283	17,163.283	0
replay	buffer	2	52,957.500	52,957.500	67,015.050	22,089.309
replay	decode	2	78,904	78,904.000	103,189.600	38,161.139
replay	feature	2	41,979.500	41,979.500	43,025.750	1,644.023
replay	processing	2	98,101	98,101.000	111,122.200	20,460.842
replay	schedule lag	2	61,372	61,372.000	64,603.000	5,077.027
replay prediction	app	2	3,164	3,164.000	3,173.900	15.556
replay prediction	native	2	1,100.286	1,100.286	1,247.450	231.246

Table 55: Real-time prediction boundaries and lag

Boundary s	Lag ms	Processing ms	Native us	App us	Rows max
100.000	57.782	112.569	1,263.802	3,175	5,453
150.000	64.962	83.633	936.771	3,153	5,489

Table 56: Real-time predicted probabilities (class order: bus, car, train)

Boundary s	Label	Bus	Car	Train
100.000	0	0.8510025740	0.0148966247	0.1341007650
150.000	0	0.9686130285	0.0228154883	0.0085715093

The run was deliberately cancelled after two boundaries. It is a pacing/cancellation check, not a completed 15-

minute trial. Decode totals in the raw record are charged to the next emitted window; early buffering measurements used integer-microsecond rounding per event.

13.5 Final committed APK smoke checks

Diagnostic results below are kept separate from the 99-run study. They do not satisfy a five-process benchmark protocol.

Model: sydney-v1-w100-s42; mode: latency; status: complete.

Source APK commit: c9b3540ac43504fa19b399e06e8b0bcc9f0bf193. Result time (UTC): 2026-10-29T06:43:05.596223Z.

Table 57: Diagnostic timing distributions (microseconds)

Operation	Metric	N	Mean	Median	p95	SD
first	app	1	3,499	3,499	3,499.000	0
first	native	1	654.271	654.271	654.271	0
load	app	1	15,455	15,455	15,455.000	0
load	native	1	10,987.345	10,987.345	10,987.345	0
warm	app	1,000	708.504	576.500	2,214.550	518.270
warm	native	1,000	179.415	175.808	229.224	26.609

Model: sydney-v1-w100-s42; mode: replay_fast_memory; status: complete.

Source APK commit: c9b3540ac43504fa19b399e06e8b0bcc9f0bf193. Result time (UTC): 2026-10-29T06:43:09.035067Z.

Table 58: Diagnostic timing distributions (microseconds)

Operation	Metric	N	Mean	Median	p95	SD
load	app	1	12,333	12,333	12,333.000	0
load	native	1	10,609.949	10,609.949	10,609.949	0
replay	buffer	15	1,994.497	1,695.366	3,361.924	699.259
replay	decode	15	23,644.467	20,402	42,244.100	10,035.020
replay	feature	15	15,145.733	14,013	19,275.300	2,640.207
replay	processing	15	22,263.364	19,053.144	38,393.741	7,822.407
replay prediction	app	15	5,123.133	2,888	17,141.400	5,842.444
replay prediction	native	15	2,501.007	888.021	12,396.512	4,400.022

13.6 Diagnostic process memory

Table 59: Total PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	132,072	132,072	132,072.000	0
100	42	loaded	1	141,532	141,532	141,532.000	0
100	42	replay end	3	175,225.667	184,741	187,844.200	19,543.286
100	42	replay observed	3	178,726.333	185,477	188,375.000	14,570.324
100	42	warmed	1	161,141	161,141	161,141.000	0

Table 60: Resident set size by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	196,832	196,832	196,832.000	0
100	42	loaded	1	206,240	206,240	206,240.000	0
100	42	replay end	3	241,726.667	251,624	254,946.800	20,423.647
100	42	replay observed	3	245,858.667	252,552	255,457.200	14,478.963

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Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	warmed	1	228,124	228,124	228,124.000	0

Table 61: Native PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	23,070	23,070	23,070.000	0
100	42	loaded	1	25,986	25,986	25,986.000	0
100	42	replay end	3	27,017.667	27,005	27,085.100	70.854
100	42	replay observed	3	26,798	26,978	27,140.000	476.235
100	42	warmed	1	26,222	26,222	26,222.000	0

Table 62: Dalvik PSS by phase (KiB; recomputed from snapshots)

Window s	Seed	Phase	N	Mean	Median	p95	SD
100	42	baseline	1	1,224	1,224	1,224.000	0
100	42	loaded	1	1,476	1,476	1,476.000	0
100	42	replay end	3	25,970.667	30,616	40,001.200	17,855.116
100	42	replay observed	3	30,724	30,968	40,349.600	10,792.069
100	42	warmed	1	19,452	19,452	19,452.000	0

Resident-set high-water values are lifetime process maxima and include earlier validation. They are not instantaneous incremental model peaks.

Table 63: Lifetime high-water RSS extrema observed (KiB)

Window s	Seed	Snapshots	Minimum	Maximum
100	42	9	198,172	276,324

14 Model configuration and provenance

One saved full-cohort pipeline (window 100 s, seed 42) was reused; eight candidates were refitted with their saved winning parameters and original training settings. No hyperparameter search was repeated. Seeds vary the selected hyperparameters as well as the stochastic fit, so these are nine fitted deployment candidates rather than a controlled seed-only ablation. Winning inner-validation scores below are selection scores, not independently held-out deployment accuracy.

Table 64: Saved winning hyperparameters

W s	Seed	colsample_bytree	gamma	learning_rate	max_depth	min_child_weight
100	42	0.9522995241	3.946944959	0.1242421157	5	9.825839391
100	43	0.8418210736	3.987275066	0.1221190926	6	9.788475479
100	44	0.9018471879	3.352170432	0.1738348954	8	1.971797555
120	42	0.802438163	3.414628846	0.02922536493	7	8.09628277
120	43	0.529412748	2.742336576	0.1837949191	7	6.678528406
120	44	0.6871631409	2.540940965	0.08366016271	6	4.071977736
140	42	0.5370192984	2.988757342	0.03253853848	5	9.009581075
140	43	0.5287690784	3.595019549	0.09711487981	4	4.435658449
140	44	0.9901504606	3.905773741	0.1122951052	5	5.343995981

Table 65: Saved winning hyperparameters

W s	Seed	n_estimators	reg_alpha	reg_lambda	subsample
100	42	300	1.458597762	6.691378894	0.8550459929
100	43	1000	0.7892498991	8.184642555	0.8348413581
100	44	800	1.820140783	7.21029362	0.8071947266
120	42	1200	0.3384516617	4.556809733	0.906869307
120	43	1100	1.810451854	9.181534415	0.9583820597
120	44	300	1.811073996	6.459546467	0.6094138924
140	42	400	1.232785257	7.05267816	0.7745191092
140	43	400	0.07766148026	7.954171263	0.8538098431
140	44	400	1.496993233	5.006418367	0.9884102245

Table 66: Saved inner selection and preprocessing state

W s	Seed	Winning trial	Inner macro F1	Indicators
100	42	81	0.8079870471	0
100	43	70	0.8088880238	0
100	44	62	0.8166742236	0
120	42	76	0.8019748442	0
120	43	22	0.8152210967	0
120	44	91	0.8071876945	0
140	42	58	0.8074015076	0
140	43	87	0.8076530742	0
140	44	90	0.8124883519	0

Table 67: Fitted training-partition medians (identical across seeds within each window)

Ordered feature	100 s	120 s	140 s
accelerometer#mean	9.852294922	9.851911545	9.852121353
accelerometer#standard_deviation	0.3885540068	0.3959730864	0.3960879743
accelerometer#range	6.060852051	6.539665699	6.967610359
accelerometer#minimum	7.076412678	6.86507225	6.750342369
gyroscope#mean	0.03870636597	0.03941862658	0.03970582783
gyroscope#standard_deviation	0.05658984557	0.06044943258	0.06151239201
gyroscope#range	0.4165867865	0.4498409927	0.4783529639
gyroscope#minimum	0.0007481313078	0.0006296637002	0.0005506120506
magnetometer#standard_deviation	5.107232571	5.486107349	5.826477051
magnetometer#range	25.84284973	28.6071682	31.27112961
magnetometer#minimum	44.28411484	43.09212494	42.0438118

Table 68: Training and export dependency versions

Item	Recorded value
joblib	1.5.3
numpy	2.4.6
onnx	1.22.0
onnxmltools	1.16.0
onnxruntime	1.24.3
pandas	3.0.3
pyarrow	24.0.0
scikit-learn	1.8.0
scipy	1.17.1
skl2onnx	1.20.0
xgboost	3.4.1

Source training run: sydney-20261007. No independent test cohort, quantization, accelerator, live-sensor, energy or background-service measurements are included. The complete five-repetition real-time study was not run.

15 Artifact integrity and reproducibility

All values come from locally saved raw exports and the prepared manifest. The source LaTeX is self-contained. The companion archive preserves full-precision aggregate CSV/JSON and per-call/per-snapshot CSVs; it avoids printing 45,000 individual timed calls in a result-summary document.

Aggregate tables display three decimal places except exact integer counts/bytes, scientific conversion errors and explicitly higher-precision model configuration. Repeated N=1 diagnostics are not pooled with the primary study or with a different app commit. Byte counts refer to the measured APK, not a later rebuild.

Manifest `cd957a4aab94ecb2ae0a7c1a6c94141e971b113028fa3033e3078162156c672f`

Measured APK `d7fc3a93f682786081b4097cfcabb582d6e02deeb623214c444ee980b6a629ff`

Summary JSON `8e3bd70c93ebbbf032d160741c501d5f6686d1f42f33ed2ff818e1d9d5629d5a`

sydney-v1-w100-s42 `88ed28211db900bbb8f24888a99ab6cdf46cc5ee565876aa02c2662918a8b2f4`

sydney-v1-w100-s43 `573a553dca81a5b2be11fe4a55d55138029cda0403e77d354d0f4e0fd69ab903`

sydney-v1-w100-s44 `94de93cb91bb74d24d02a0f7ec2c8a1a1757317b087ca0523809fe43a833a9c5`

sydney-v1-w120-s42 `bca510da164198034d76eea0ef1fcf3a0bfa553b46f6a5594a1edefe4ad0f78e`

sydney-v1-w120-s43 `99e38c62626962cc38dca3a674197e22198620059849a9ca3f14e52f99976997`

sydney-v1-w120-s44 `2b1b0f1325338e4f02e3debe6e485b6b34f6c2a91ccf90a75065f2f818354883`

sydney-v1-w140-s42 `85be45f46e4bde9f68371828a2110fd21dc093ad9ace418dbd0b10f270c302b5`

sydney-v1-w140-s43 `6feb3baa3247e09dcc1b378d818da178dacb5695bf2f07ac51ddbf960a64a3c0`

sydney-v1-w140-s44 `47f52e3599a93bd27a91b910bf8a721b20e117c66eb4b4841657a70f2f089512`